Survival Analysis Report
Landmark Models for Early Prediction of Severely Impaired RAP

Introduction

Continuous monitoring of cerebral physiology is essential in managing traumatic brain injury (TBI). The compensatory reserve index (RAP) reflects the brain's ability to accommodate intracranial changes, and its impairment may precede clinical deterioration. However, most studies focus on descriptive analysis rather than modeling when such deterioration occurs.

This analysis applies a time-to-event framework to model the onset of sustained severe compensatory reserve failure (i.e., RAP<0). A landmark-based design was used, where features from the early monitoring period were used to predict future deterioration among patients who remained event-free during that window.

Kaplan–Meier analysis, Cox proportional hazards models, and Random Survival Forests were employed to evaluate both interpretability and predictive performance. The objective was to determine whether early RAP and multimodal physiological features can predict the timing of severe deterioration and to compare classical and machine learning survival approaches in this setting.

Dataset

The Winnipeg Acute TBI dataset was utilized in this analysis.

Methods

- Landmark windows: 3 hours and 6 hours. They were chosen to balance early prediction with data reliability.
- Event detection: first sustained RAP value below -0.1 for 10 consecutive minutes. A strict threshold at 0 is sensitive to noise and small fluctuations. Using -0.1 ensures the drop is clearly below the boundary, capturing true severe deterioration rather than borderline or transient changes.
- Patients with an event during or before the landmark window were excluded from that landmark analysis.
- Patients with monitoring that did not extend beyond the landmark were also excluded.
- Features were computed from the early window only. These included RAP, ICP, CPP, MAP, and AMP summaries; separate RAP-only models also used RAP variability and burden features.
- Outcome: time from landmark to first sustained event, or time from landmark to end of follow-up if censored.
- Evaluation used log-rank tests for Kaplan-Meier analyses and concordance index (C-index) for predictive performance.

Privacy-Aware Preprocessing Extension

A privacy-aware preprocessing layer was added before feature extraction and survival modeling. This layer was designed to reduce re-identification risk while preserving the temporal and physiological information required for landmark survival analysis.

Input field	Privacy action	Modeling role after transformation
ID / file-derived patient ID	Hashed into anon_patient_id using deterministic SHA-256 hashing, then raw ID removed	Allows patient-level grouping without exposing original ID
Name	Removed if present	Not used for modeling
Age	Converted into age_group bins: 18-29, 30-39, 40-49, 50-59, 60-69, 70+	Retains demographic signal while removing exact age
Sex (M/F)	Standardized to sex_group	Used as a privacy-safe categorical covariate
DateTime	Converted to time_minutes from each patient's first timestamp, then absolute DateTime removed	Preserves temporal ordering without exposing calendar time
Physiological signals	Retained after de-identification	Used for early-window feature extraction

This design should be described as privacy-aware pseudonymization and de-identification, not as full anonymization. The main privacy gain is that direct identifiers, raw patient IDs, exact ages, and absolute timestamps are not retained in the modeling dataset.

Privacy Processing Audit

Privacy audit item	Result
Demographic rows loaded	113
Signal files processed	109
Files with demographic match	109
Files with successful time conversion	109
Files with age binning	109
Files with sex standardization	109
Direct identifier columns removed	0 files in current signal files because name was not present in the signal CSVs
Raw patient IDs retained in modeling dataset	No; replaced by anon_patient_id

Cohort Feasibility by Landmark

Landmark	Included	Early-event excluded	Short follow-up excluded	Later events	Censored
3 hours	73	29	7	49	24
6 hours	66	38	5	42	24

Interpretation: the 3-hour landmark preserved a larger analyzable cohort and more events. That generally improves model stability and partially explains why the 3-hour models were more interpretable and more competitive.

Kaplan-Meier Results

Grouping variable	3-hour log-rank p	6-hour log-rank p	Interpretation
Median split of early RAP mean	0.0016	0.0283	Significant at both landmarks; stronger separation at 3h
Median split of early ICP mean	0.3531	0.2122	Not significant at either landmark

Clinical meaning: Patients with lower early RAP values experienced earlier transition into sustained severe RAP deterioration. In contrast, average early ICP alone did not separate risk groups well, suggesting that compensatory reserve status captured by RAP is more informative than ICP burden alone for this endpoint.

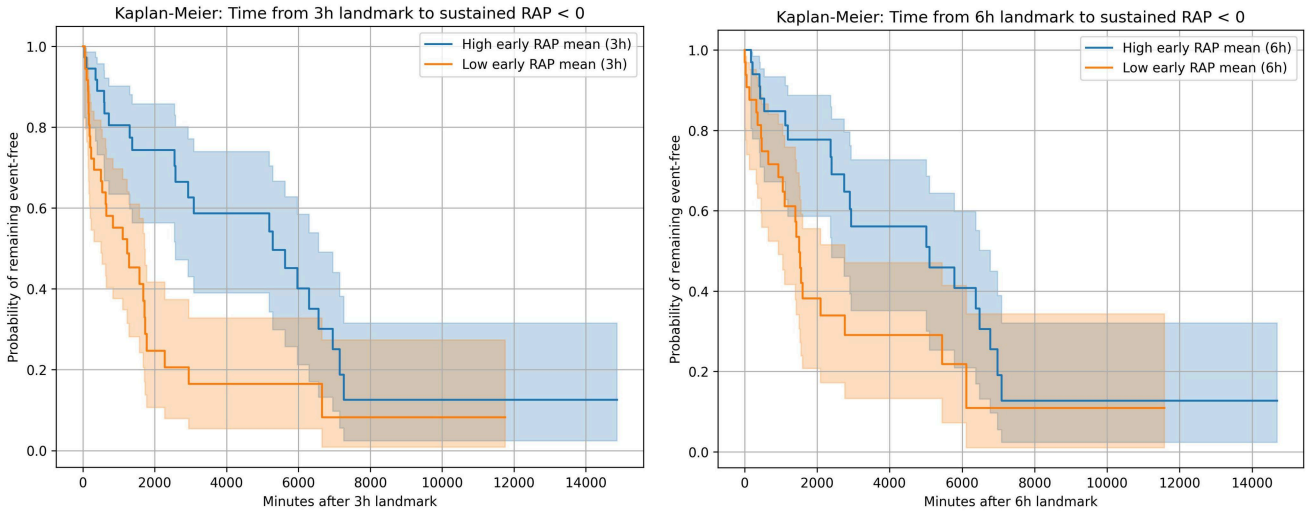

Figure: Kaplan–Meier Survival Curves Comparing Time to Sustained RAP Deterioration by Early RAP Stratification at 3-hour and 6-hour Landmarks

Both plots above show Kaplan–Meier curves comparing high vs low early RAP mean, but at different landmark times (3h vs 6h).

- In both cases, higher early RAP is associated with better outcomes, as the blue curves remain higher, indicating a greater probability of remaining event-free.
- The separation between groups is more pronounced at 3 hours, with earlier and clearer divergence between high and low RAP groups.
- At 6 hours, the separation is still present but weaker, suggesting some loss of discriminative signal or reduced cohort size.
- The wider confidence intervals at later times in both plots reflect increasing uncertainty due to fewer patients at risk.

Overall, the 3-hour landmark provides stronger early risk stratification, while the 6-hour model shows similar trends but with reduced clarity.

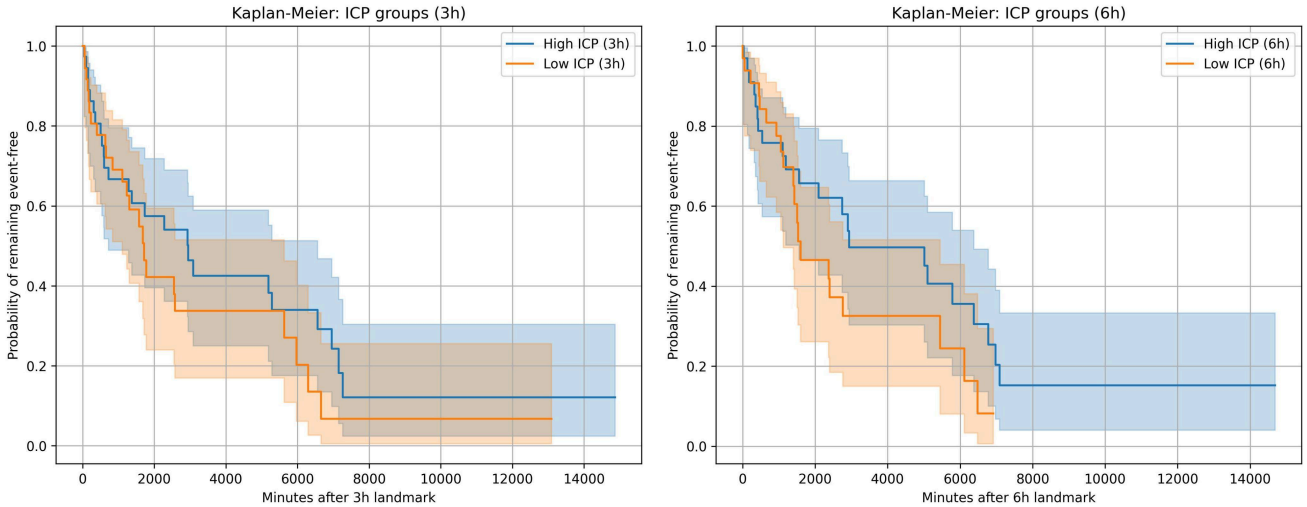

Figure: Kaplan–Meier Survival Curves Comparing Time to Sustained RAP Deterioration by Early ICP Groups at 3-hour and 6-hour Landmarks

In the previous figure, patients were stratified based on early RAP values, while in this figure, stratification is based on early ICP values at both 3-hour and 6-hour landmarks.

For the RAP-based curves, there is a clear separation between the groups. Patients with higher early RAP maintain a higher probability of remaining event-free, while those with lower RAP deteriorate earlier. This separation is especially pronounced at the 3-hour landmark and remains visible at 6 hours.

In contrast, the ICP-based curves show much weaker separation. The high and low ICP groups follow very similar trajectories, with considerable overlap between the confidence intervals. This indicates that early ICP alone does not effectively distinguish between patients who deteriorate earlier versus later.

Comparing the two figures, RAP provides a stronger and more consistent stratification of risk than ICP. This suggests that compensatory reserve, as captured by RAP, is more informative for predicting the timing of deterioration than pressure-based measures alone.

Cox Proportional Hazards Models

Model summary table

Model	n	Events	Key significant findings	PH assumption	Convergence	C-index
3h, all physiologic means	70	47	rap_mean_3h HR 0.028, p = 0.00085; others not significant	Looks okay	No warning	0.687
3h, RAP-only	73	49	rap_mean_3h HR 0.054, p = 0.0226	Violations for rap_mean_3h and pct_rap_gt_0_4_3h	No warning	0.673
6h, all physiologic means	64	41	icp_mean_6h, cpp_mean_6	Severe violations	Convergence warning	0.531

			h, map_mean_6h significant; rap_mean_6h not significant			
6h, RAP-only	66	42	rap_mean_6h HR 0.045, p = 0.0216	Violation for pct_rap_gt_0_4_6h	No warning	0.649

Interpretation of Cox findings

- The most consistent signal in the Cox analyses was early RAP mean. Hazard ratios far below 1 at both landmarks indicate that higher early RAP was protective, meaning patients with better early compensatory reserve tended to deteriorate later.
- The 3-hour multivariable Cox model is the most credible Cox model in this analysis. It had the best Cox discrimination, a strong and statistically significant RAP effect, and no major proportional hazards problems.
- The 6-hour multivariable Cox model should not be treated as clinically interpretable in its current form. It produced a convergence warning and showed extreme proportional hazards violations. More importantly, ICP, CPP, and MAP were entered together even though CPP is derived from MAP and ICP, which creates structural collinearity and can distort effect directions and significance.
- The RAP-only Cox models were cleaner. They preserved the central biological message that lower early RAP predicts earlier severe deterioration, although some proportional hazards issues remained for burden variables.

Random Survival Forest Results

Model	Test C-index	Top positive importance features	Negative or unstable importances	Take-home
3h, all physiologic means	0.613	rap_mean_3h, cpp_mean_3h, map_mean_3h	icp_mean_3h, amp_mean_3h	Moderate performance; RAP still leads
3h, RAP-only	0.696	pct_rap_0_to_0_4_3h, rap_mean_3h, pct_rap_gt_0_4_3h	rap_std_3h	Best apparent test performance
6h, all physiologic means	0.669	rap_mean_6h, map_mean_6h, amp_mean_6h	icp_mean_6h	Best 6h model; suggests nonlinear value in longer window
6h, RAP-only	0.615	rap_mean_6h, pct_rap_gt_0_4_6h	pct_rap_0_to_0_4_6h, rap_std_6h	Weaker than 6h all-variable RSF

Interpretation: RAP remained the dominant variable in the RSF models as well. The strong 3-hour RAP-only RSF suggests that non-linear relationships within RAP-state burden may contain useful signal. However, permutation importances that become negative usually indicate instability, noise, or strong redundancy between predictors. Because these RSF results came from a single train-test split on a small cohort, they should be viewed as preliminary rather than definitive.

Clinical Interpretation

- Early RAP appears to be a better marker of near-term severe reserve failure than mean ICP alone. This is clinically plausible because RAP is intended to reflect compensatory reserve status, whereas ICP alone may remain relatively similar across patients before overt decompensation appears.
- The persistence of the RAP signal across both landmarks supports the idea that early reserve impairment precedes later severe deterioration. In practical terms, patients entering the monitoring period with lower RAP may merit closer observation or earlier intervention.
- The 3-hour results suggest that clinically useful prognostic information is already present early in the monitoring period. This is valuable because an earlier landmark offers more time for risk stratification and potential treatment adaptation.
- The RSF finding that time spent in the borderline RAP range (0 to 0.4) contributed strongly at 3 hours suggests that transitional physiology may matter. Patients who spend more time near the boundary between adequate and impaired reserve may be unstable even if their mean RAP is not frankly low.

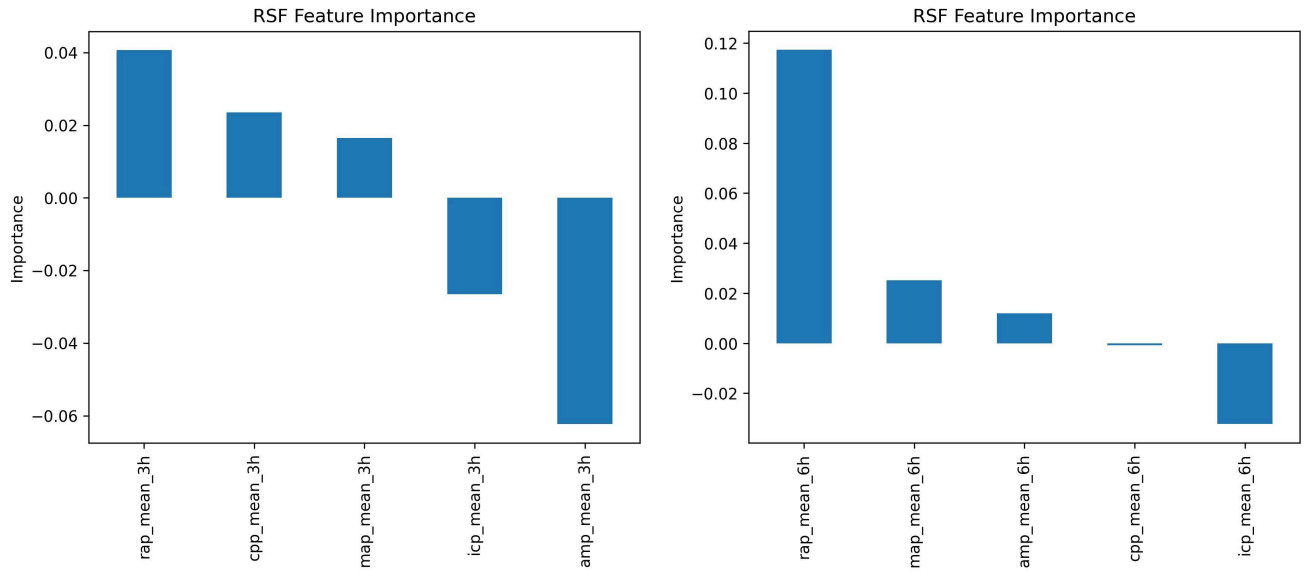

Figure: Random Survival Forest Feature Importance for Early Physiological Predictors at 3-hour and 6-hour Landmarks

This figure shows the relative importance of early physiological features in predicting time to sustained RAP deterioration using Random Survival Forest models at 3-hour and 6-hour landmarks.

Across both models, early RAP mean is the most influential predictor, with a stronger contribution at 6 hours. Other variables such as MAP and CPP show moderate importance, while ICP and AMP contribute less or show negative importance, suggesting limited or unstable predictive value.

Model Comparison and Preferred Framing

- For interpretability and statistical defensibility, the preferred primary model is the 3-hour Cox model with carefully selected predictors, especially RAP-based measures.
- For predictive performance, the strongest observed result was the 3-hour RAP-only RSF (test C-index 0.696), but that advantage over the 3-hour Cox model was small and needs confirmation with repeated resampling or cross-validation.

- The 6-hour landmark is not useless, but it is less attractive as the main analysis because it excludes more early deteriorators and leaves fewer events for modeling. It may still be useful as a sensitivity analysis.
- Taken together, these findings support a simple story: an earlier landmark is more efficient, and RAP is the most clinically meaningful predictor family for this endpoint.

Privacy-Aware 3-Hour Landmark Results

The privacy-aware notebook was then run for the 3-hour landmark using de-identified physiological features and privacy-safe demographic variables.

Metric	Privacy-aware 3-hour result
Included patients	78
Excluded early events	29
Excluded short follow-up	2
Later events after landmark	52
Censored patients	26
Median follow-up after 3h	1480.68 minutes
Kaplan-Meier RAP log-rank p-value	0.00147
Cox model C-index	0.719
RSF test C-index	0.606

Privacy-aware Cox variable	Hazard ratio	p-value	Interpretation
rap_mean_3h	0.021	0.0010	Higher early RAP remained strongly protective
icp_mean_3h	1.067	0.1748	Not significant
cpp_mean_3h	1.041	0.0845	Borderline but not conventionally significant
map_mean_3h	0.969	0.2864	Not significant
amp_mean_3h	0.976	0.8976	Not significant
age_group_70+	5.763	0.0054	Older age group showed higher hazard in this model
sex_group_Male	1.985	0.1249	Not significant

The privacy-aware Cox model preserved the central finding from the original analysis: early RAP mean remained the dominant and statistically significant predictor. Adding privacy-safe demographic variables did not weaken the RAP signal; instead, the 3-hour Cox C-index increased from 0.687 in the original physiologic-only model to 0.719 in the privacy-aware model. This comparison should be interpreted cautiously because the privacy-aware model also included age and sex and had a slightly different analyzable cohort.

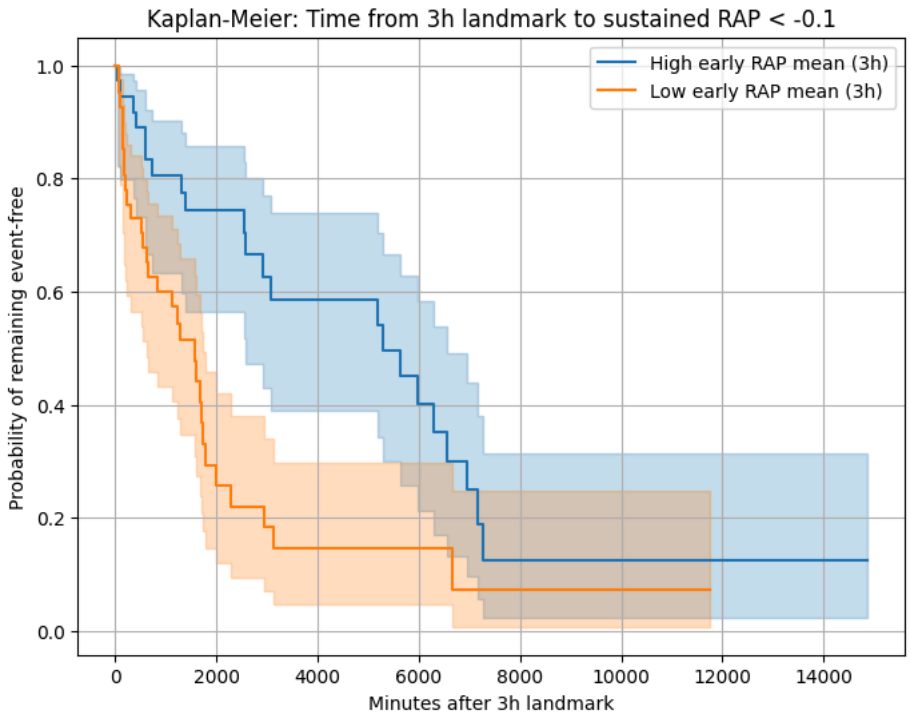

Figure: Privacy-aware 3-hour Kaplan-Meier curve using de-identified data and early RAP mean stratification.

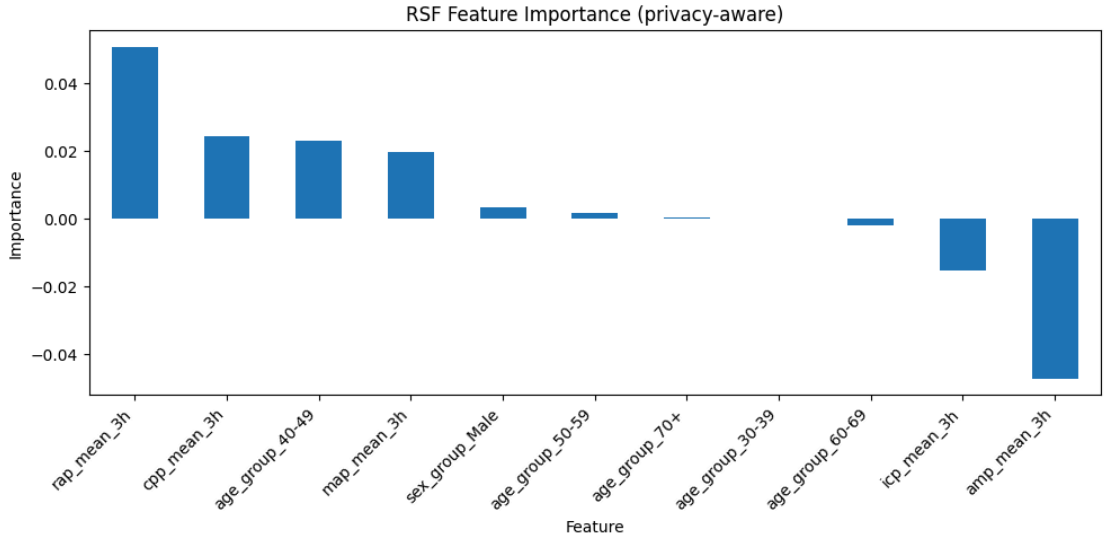

Figure: Privacy-aware 3-hour Random Survival Forest permutation feature importance including physiological and privacy-safe demographic variables.

Feature	Permutation importance
rap_mean_3h	0.0506
cpp_mean_3h	0.0245
age_group_40-49	0.0229
map_mean_3h	0.0197
sex_group_Male	0.0035

age_group_50-59	0.0016
age_group_70+	0.0003
age_group_30-39	0.0000
age_group_60-69	-0.0019
icp_mean_3h	-0.0152
amp_mean_3h	-0.0471

In the privacy-aware RSF model, RAP mean remained the top-ranked predictor. CPP and MAP also contributed positively, while ICP and AMP had negative permutation importance, suggesting instability or redundancy in this small test split. The RSF C-index was similar to the original 3-hour physiologic RSF model and should be treated as preliminary.

Privacy-Utility Interpretation

Analysis setting	Main privacy state	KM RAP p-value	Cox C-index	RSF C-index	Interpretation
Original 3h	No formal privacy layer	0.0016	0.687	0.613	Strong RAP signal; baseline utility
Privacy-aware 3h	Hashed ID, age bins, standardized sex, relative time only	0.00147	0.719	0.606	RAP signal retained after de-identification

The main privacy-utility finding is that the clinically meaningful RAP signal was retained after privacy-preserving preprocessing. The Kaplan-Meier separation remained statistically significant, the Cox model continued to identify early RAP mean as strongly protective, and the RSF model still ranked RAP mean as the most important feature. Therefore, the project can be framed as a privacy-aware survival modeling workflow rather than only a conventional survival analysis.

Interpretation

- The privacy-aware extension suggests that patient-level risk modeling can still retain useful signal after removing raw identifiers, exact ages, and absolute timestamps.

Conclusion

This analysis demonstrates that early RAP-derived features provide meaningful predictive information regarding the timing of severe compensatory reserve deterioration. Across all approaches, lower early RAP was consistently associated with earlier deterioration.

Among the tested configurations, the 3-hour landmark offered the best balance between sample retention, event yield, interpretability, and predictive performance. Both Kaplan–Meier and Cox models showed clear and consistent stratification based on RAP, while ICP-based stratification demonstrated limited discriminative value.

Random Survival Forest models further supported the dominant role of RAP, although their performance advantage over Cox models was modest. Overall, these findings suggest that early compensatory reserve status, as captured by RAP, is more informative than pressure-based measures alone for anticipating near-term deterioration.

The added privacy-aware pipeline shows that the same clinical modeling task can be performed after applying practical privacy protections: hashed patient identifiers, removal of raw IDs and names, age binning, standardized sex categories, and conversion of absolute timestamps into relative follow-up time.